



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Mechanical Engineering
In-Semester Examination II
Academic Year: 2025-26 (Semester I)

Class: TY	Subject: Machining and Metrology	Subject Code: PCC-ME307
Date: 15-Oct-2025	Time: 11:00 to 11:45	Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.

Q.N.	Solve all the questions	Marks	CO	BT
Q.1	Describe the evolution of cutting tool materials from early developments to modern advancements. Present the key milestones in a chronological timeline.	5	CO5	BT1
Q.2	Why is high velocity desired in grinding?	5	CO2	BT4
Q.3	If in turning of a steel rod by a given cutting tool (material and geometry) at a given machining condition (s_0 and t) under a given environment (cutting fluid application), the tool life decreases from 80 min to 20 min. due to increase in cutting velocity, V_C from 60 m/min to 120 m/min., then at what cutting velocity the life of that tool under the same condition and environment will be 40 min.?	5	CO3	BT3
Q.4	Estimate the machining time that will be required to finish a vertical flat surface of length 100 mm and depth 20 mm by an 8 teeth HSS end mill cutter of 32 mm diameter and 60 mm length in a milling machine. Assume, $V_C = 30$ m/min, $s_0 = 0.12$ mm/tooth.	5	CO4	BT3

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Demonstrate the fundamentals of the metal cutting process and select appropriate cutting tools for various machining operations.
CO 2	Exhibit knowledge of conventional and finishing machining processes, along with their operational principles and applications.
CO 3	Explain characteristic terminologies related to cutting tools and illustrate the concepts essential to efficient machining practices.
CO 4	Calculate machining parameters such as machining time, depth of cut, and feed rate for different machining scenarios.
CO 5	Select suitable machining processes for specific products, ensuring alignment with design intent, material properties, and manufacturing constraints.
CO 6	Use appropriate measuring instruments to accurately measure linear, angular dimensions, surface finish, gear parameters, and determine limits, fits, and tolerances.



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded

Academic Year: 2025-26

Semester: I

Examination: ISE-II

Class: Third Year

Branch: Mechanical Engineering

Date: 14/10/2025

Time: 11:00 - 11:45

Subject: Machine Design - I

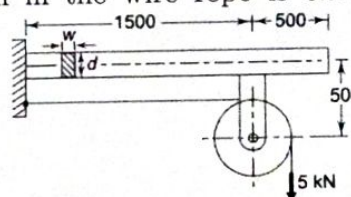
Code: PCC-ME305

Maximum Marks: 20

General Instructions:

- I. This paper contains 1 (One) page. Figures to right indicate full marks.
- II. Draw neat & compact sketches wherever necessary.
- III. Assume data wherever required and state them clearly.

Que. No.	Question	Marks	CO	BT
1.	Compare the static and varying stresses. Mention different types of varying stress with daily life simply observable example.	5	CO2 CO3	BT4
2.	Two rods are connected by means of a cotter joint. The rods are subjected to a tensile force of $50kN$. The cotter is made of steel 30C8 ($S_{yt} = 400N/mm^2$) and the factor of safety is 4. The width of the cotter is five times of thickness. Evaluate width and thickness of the cotter.	5	CO2	BT5
3.	A cantilever beam of rectangular cross-section is used to support a pulley as shown in figure. The tension in the wire rope is $5kN$. The beam is made of cast iron $S_{ut} = 200N/mm^2$ and the factor of safety is 2.5. The ratio of depth to width of the cross-section is 2. Determine the dimensions of the cross-section of the beam.	5	CO2	BT5
4.	What is stress concentration and stress concentration factor. Explain possible ways of reducing its severity in order to avoid the chances of failure in fluctuating loading.	5	CO3	BT1 BT2



**** ALL THE BEST ****

- CO1:** Analyze the fundamental principles of machine design to identify and evaluate the design requirements, material selection, and safety considerations for various mechanical components.
- CO2:** Apply theories of static failure to design and analyze mechanical components subjected to static loading.
- CO3:** Evaluate and design mechanical components to withstand fluctuating and fatigue loading conditions using endurance limit estimation.
- CO4:** Design and analyze shafts, keys, and couplings as per industrial standards.
- CO5:** Develop designs for threaded and welded joints subjected to static, eccentric, and fluctuating loads.

BT1: Remember, **BT2:** Understand, **BT3:** Apply, **BT4:** Analyze, **BT5:** Evaluate, **BT6:** Create.



Date: 14-Oct-25

Time: 14:30-15:15

- Assume suitable data, if necessary.
- Solve all questions.
- Figures to the right indicate full marks.
- Allow nonprogrammable calculator.

Q.N.	Answer the following questions.	Marks	CO	BT
Q.1	Does the cooling time and the temperature at which cooling is performed affect the microstructural changes in the material? Justify your answer with an example and a diagram	5	CO2	BT2
Q.2	How to classify the polymer? Based on various groups	5	CO1	BT1
Q.3	What is the Annealing process? Explain its types and applications.	5	CO3	BT1
Q.4	How would you design a heat treatment sequence for the <i>Chhatrapati Shivaji Maharaj Sword</i> to achieve hardness at the edge but toughness at the core?	5	CO3	BT4

Course Outcomes (CO)

- CO1. Identify the various materials, viz., metals, polymers, ceramics, and composites.
- CO2. Analyse the dependance of mechanical properties on the microstructure.
- CO3. Select appropriate heat treatment for metals and alloys for particular application.
- CO4. Acquire knowledge about powder metallurgy process and its applications.
- CO5. Acquire knowledge about advanced materials and its applications.

Bloom's Taxonomy: BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

~~ Best of luck ~~